SUPPORTING ADULT GUIDANCE

A table can predict a line

Straight Lines and Connected Representations. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner **generates coordinates from a linear rule** and discovers the **constant first difference** — the fixed amount the y values change by each step. The emphasis is on **prediction**: seeing that a few values reveal the whole line, so the graph can be predicted before it is drawn, and that a constant step is exactly what distinguishes a straight line from other patterns.

Driving Question: *How can a few coordinate pairs reveal a whole linear pattern?*

The mathematics, in plain terms

Given a rule like $y = 3x + 2$, you make a table by **substituting** each x: when $x = 0$, $y = 2$; when $x = 1$, $y = 5$; when $x = 2$, $y = 8$, and so on. Written out, the y values are 2, 5, 8, 11 — going up by **3** every time. That fixed jump is the **first difference**, and when it is the **same** every step the pattern is a **straight line**; the size of the step is the gradient m, and the first value (at $x = 0$) is c. Because the step never changes, you do not need to plot every point — from a couple of rows you can predict the direction (up if the step is positive, down if negative), the value when $x = 0$, and any future value (keep adding the step). This also gives a quick test of whether something is linear at all: if the first differences change — as with 1, 4, 9, 16 (steps 3, 5, 7, the square numbers) or 1, 2, 4, 8 (doubling) — the pattern is **not** a straight line.

A few rows contain the whole line

The heart of this lesson is prediction from a constant rate: a straight-line table is 'start, then add the same step, again and again', so a few values already fix the whole line. The most valuable habit to protect is **predict before you plot** — asking which way the line goes and where it starts, and only then revealing the graph to confirm. That turns the graph from something to copy into something to anticipate. Keep 'first difference' in plain language ('the step') at first. The non-linear examples are important, not a distraction: meeting a pattern whose step changes is what makes 'constant rate' mean something. This lesson leads straight into the next two, which name and isolate the two numbers — the intercept c and the gradient m.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Table-to-Line Predictor: fill a table from a rule, spot the constant step, and predict the line before revealing the graph.	~15 min

Movement	Step-to-Build: grow the table one row at a time and watch the line build from the constant step.	~8 min
Reflection	First-Difference Detective: decide whether each table's step is constant — a straight line or not.	~10 min
Creativity	Predict a Mystery: work out a whole line from just its first two rows.	~5 min
Rest	A pause after generating values and spotting a pattern.	~4 min
Nutrition	Contextual (Mode A): steady rates that let you predict ahead — bills, distances, savings.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Working out y values but not noticing the constant step

Repair: “Look down the y column — how much does it change each time? Is that jump always the same?”

Assuming every number pattern is a straight line

Repair: “Check the steps: 1, 4, 9, 16 — are the jumps equal? If the step changes, is it still a straight line?”

Adding the step to x instead of to y

Repair: “The constant step is how much y changes each time x goes up by 1. Which column gets the step added?”

Thinking you must plot every point to know the line

Repair: “If the step never changes, do you need every row — or can you jump ahead by adding the step?”

Reading a falling line (negative step) as ‘no pattern’

Repair: “The y values go down by the same amount each time. Is a constant negative step still a straight line?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: substituting a value into a rule such as $y = 3x + 2$, and reading and plotting coordinates. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the prediction — and the constant step — with the learner:

- “Use the rule to find y for x = 0, then x = 1. What changed?”
- “Is that step the same again for the next row? So is it a straight line?”
- “Before we draw it — does the line go up or down, and where does it start?”
- “How could you find y at x = 10 without filling every row in between?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Is ‘first difference’ the same as gradient?”

Yes — for a straight line the first difference (the constant step in y) is exactly the gradient m. The formal name and notation come in the next lesson; here ‘step’ is enough.

“Why ask them to predict before showing the graph?”

Because the whole point is that a table already predicts the line. Revealing the graph first removes the thinking; predicting first, then checking, is where the learning happens.

“Why include patterns that are not straight lines?”

To make ‘constant rate’ meaningful. A learner who only ever sees linear tables cannot really tell what makes a line a line; one changing-step example fixes that.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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